

Implementation Guide

Implementing an Auditable Grid-Asset Evidence Foundation

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September 2026

OGARD Implementation Guide

Version 0.1.2

Deniz Bektas authored this guide using original work, synthetic examples, and public sources.

Companion papers: OGARD Technical Whitepaper; OGARD Architecture and Data Model Specification; OGARD Synthetic Benchmark Methodology and Validation Plan.

Versioning: This paper is version 0.1.2. The methodology, model profile, and contradiction taxonomy it describes are version 0.1. The executable reference software has its own release numbers (Executable Reference Release v0.2.1 and later).

Repository: <https://github.com/DenBek/ogard>

Archive: <https://doi.org/10.5281/zenodo.22929401>

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Reference implementation license: Apache License 2.0 for original repository code; see LICENSE.md.

How to cite: Bektas, D. (2026). OGARD Implementation Guide: Implementing an Auditable Grid-Asset Evidence Foundation. Version 0.1.2. <https://github.com/DenBek/ogard>

Abstract

This guide translates the OGARD technical framework into a controlled implementation lifecycle for electricity-network asset data. It is designed for bounded pilots and future production implementations in which maintenance records, asset registers, telemetry references, fault and outage information, inspection evidence, and technical documentation must be reconciled without erasing uncertainty or provenance. The guide defines implementation outputs, governance roles, readiness criteria, source contracts, identifier-regime mapping, canonical asset-and-evidence records, reconciliation rules, human review, contradiction handling, operator-governed derived status, cybersecurity controls, testing, measurement, and production-readiness gates. The recommended approach begins with one asset class and a small set of source systems, uses deterministic rules before similarity scoring, routes uncertain or consequential cases to authorized review, and keeps optional artificial-intelligence assistance advisory. Public OGARD materials remain separate from operator-controlled data and implementation records. Operator adoption, live performance, and formal conformance require deployment-specific evidence and assessment.

Keywords: grid asset data; implementation lifecycle; data provenance; entity resolution; human review; utility data governance; reconciliation; operational technology; auditability; production readiness.

How to use this guide

The guide is organized as a phase-gated implementation sequence. A team should begin with Sections 1 through 6 to define authority, scope, readiness, and pilot success criteria. Sections 7 through 14 describe the technical implementation. Sections 15 through 18 define security, testing, measurement, and production readiness. Section 19 states scope and evaluation limits. Appendices provide minimum records, phase-gate checklists, an example bounded pilot, and a glossary that can be adapted to an operator environment.

Implementation rule: Do not begin ingestion merely because source access is technically possible. Begin only after source ownership, intended use, security classification, data handling, pilot scope, and review authority are documented.

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Executive quick start

An OGARD implementation is a governed method for reconciling fragmented records about physical grid assets. It does not begin with a dashboard or an artificial-intelligence assistant. It begins with authority, scope, source evidence, identity definitions, and a review process.

A minimum pilot should focus on one asset class, two to four source systems, a limited time window, and a defined group of operational and data subject-matter experts. The implementation should preserve raw source records, separate functional location from physical equipment and telemetry identity, apply deterministic rules first, use reproducible similarity scoring only where needed, and route unresolved or consequential cases to authorized review.

Table 1. Minimum viable implementation outputs.

Output	Minimum purpose
Approved pilot charter	Defines business question, asset class, source systems, boundaries, owners, decision rights, and evaluation boundaries.
Source inventory and contracts	Records ownership, access, identifiers, time fields, quality state, security classification, and extraction expectations.
Identifier-regime map	Separates functional location, physical equipment identity, and telemetry or instrumentation identity.
Canonical asset-and-evidence model	Represents the governed asset context while preserving source records and time validity.
Versioned rule library	Defines deterministic rules, similarity methods, thresholds, owners, tests, and effective dates.
Human review queue	Presents evidence, candidate relationships, scores, contradictions, and authorized decisions.
Contradiction register	Preserves typed disagreements instead of silently selecting one source.
Derived-status rule set	Defines operator-approved status semantics, evidence requirements, review conditions, and expiry.
Audit and lineage records	Reconstructs how a source record contributed to a mapping, contradiction, or output.
Pilot measurement report	Documents scope, baselines, results, unresolved issues, limitations, and next actions.

Success principle: A pilot is successful when it produces trustworthy implementation records and exposes where evidence remains insufficient. A high automation rate is not, by itself, the objective.

1. Purpose, audience, and scope

1.1. Purpose

This guide translates the OGARD Technical Whitepaper [1] into an implementation sequence and should be read with the companion OGARD Architecture and Data Model Specification [11] and OGARD Synthetic Benchmark Methodology and Validation Plan [12]. It defines minimum artifacts, roles, controls, and phase gates for a bounded pilot and for any later production consideration. It is intended to prevent a technically plausible integration from being represented as governed merely because data has been joined successfully.

1.2. Intended audience

- Utility data engineers, platform architects, and data-product teams.
- Asset-management, maintenance, reliability, and work-management specialists.
- Supervisory control and data acquisition (SCADA) and historian specialists responsible for tags, telemetry context, and source access.
- Data governance, cybersecurity, privacy, legal, records-management, and audit teams.
- Platform operations and site-reliability engineering (SRE) teams.
- Vendors and consultants implementing operator-approved asset-data solutions.

1.3. Scope

The scope of this guide is electricity-network asset evidence. A pilot may include transformers, breakers, substations, feeders, power-electronic equipment, or another operator-selected class, with suitability evaluated for that defined scope. The direct implementation outputs are asset-evidence views, mappings, contradictions, review records, audit records, and operator-governed derived status. Any operational use remains subject to operator approval.

Implementation scope. OGARD provides a data-engineering and governance method. The operator governs system authorization, cybersecurity, safety, status definitions, decision rights, and use of outputs in maintenance or operations.

1.4. Out of scope

- Direct control of operational technology or automatic writeback to SCADA, protection, outage-management, or control systems.
- Autonomous fault diagnosis, outage prediction, predictive maintenance, or control-room decision making.
- Formal IEC Common Information Model (CIM) profile conformance, North American Electric Reliability Corporation Critical Infrastructure Protection (NERC CIP) compliance, or regulatory approval.
- A claim that one source system, algorithm, or model is universally authoritative.
- Publication of protected operator, employer, client, or vendor information.

1.5. Conventions

In this guide, must and required indicate a minimum control or artifact for an OGARD implementation claim; should and recommended indicate the default approach unless the operator documents a justified alternative; may and optional indicate a context-dependent capability. These labels define requirements within this guide; formal standards and certification use their applicable assessment processes.

2. Required implementation outcomes

The implementation should be managed around outputs that can be inspected and approved. Technology selection is secondary. A pilot that uses sophisticated software but cannot show source ownership, mapping decisions, review authority, or traceable evidence is not an OGARD implementation in the intended sense.

2.1. Evidence outcome

Every consumed output should be traceable to preserved source records, transformations, rules, scores, time context, and review actions. The implementation should distinguish original evidence from normalized values and derived outputs. W3C PROV-O provides a general model for representing entities, activities, agents, and provenance relationships [4], although OGARD does not require a specific provenance technology.

2.2. Identity outcome

The implementation should represent functional location, physical equipment identity, and telemetry or instrumentation identity as separate concepts. It should maintain time-valid relationships among them and avoid assuming that a field called `asset_id` has the same meaning in every source.

2.3. Decision outcome

Mappings and derived statuses should be accepted only through documented rules or authorized review. Uncertainty should remain visible. The output should explain what is known, what is inferred, what is disputed, and who approved any material decision.

2.4. Operational-boundary outcome

The pilot should remain within an operator-approved data environment and should not create an uncontrolled path into operational technology (OT). NIST SP 800-82 Rev. 3 emphasizes that OT security must account for performance, reliability, and safety requirements [6]. OGARD therefore treats OT source access, connectivity, and writeback as operator-governed security decisions, not default implementation features.

3. Implementation control principles

1. **Preserve evidence before interpretation.** Capture source records and metadata before normalization, matching, or status derivation.
2. **Separate assets from evidence.** A governed asset view must not erase the records that support or challenge it.
3. **Separate identity regimes.** Functional role, installed equipment, and telemetry identity have different meanings and time behavior.
4. **Route uncertainty rather than hiding it.** Ambiguous, stale, incomplete, or contradictory evidence should be reviewed or marked Unknown.
5. **Keep operator authority explicit.** Operators define state semantics, thresholds, review roles, and acceptable use.
6. **Prefer deterministic and explainable methods.** Similarity or AI assistance should be introduced only where they add controlled value.
7. **Make every material decision auditable.** Rules, scores, thresholds, actors, timestamps, and source references should be recorded.
8. **Design security and confidentiality at the start.** Public artifacts and operator-controlled data must remain separated.

9. **Measure the correct evidence class.** Synthetic metrics, pilot metrics, and production outcomes must not be combined.
10. **Start small enough to learn.** The first pilot should expose governance and identity problems without creating enterprise-wide risk.

4. Phase-gated adoption lifecycle

Implementation should proceed through explicit gates. A team may iterate within a phase, but it should not use technical progress to bypass unresolved authority, security, or evidence questions.

Table 2. Phase-gated implementation lifecycle.

Phase	Primary work	Required exit evidence
0. Authority and charter	Confirm sponsor, operator authority, problem statement, protected-data boundary, and intended use.	Approved charter; named owners; initial risk and confidentiality review.
1. Readiness and scope	Select asset class, source systems, time window, users, success criteria, and exclusions.	Readiness assessment; go/no-go decision; bounded pilot scope.
2. Source discovery	Inventory sources, identifiers, timestamps, evidence classes, access, and quality limitations.	Source inventory; data contracts; source-owner approvals.
3. Raw and conformance foundation	Ingest evidence, preserve originals, normalize controlled fields, quarantine invalid records.	Traceable raw and conformed records; validation reports; replay procedure.
4. Canonical and identity model	Implement asset, evidence, time-valid relationships, and selected CIM mapping guidance.	Approved schema; identifier-regime map; sample lineage walkthrough.
5. Reconciliation and review	Implement blocking, deterministic rules, similarity scoring, thresholds, review, and audit.	Versioned rules; calibration record; functioning review queue; decision logs.
6. Contradiction and status outputs	Implement contradiction types and operator-approved derived-status rules.	Contradiction register; rule approvals; safe Unknown behavior; expiry rules.
7. Pilot measurement and acceptance	Run the bounded pilot, measure outputs, analyze errors, and document limitations.	Pilot report; unresolved issue log; acceptance or remediation decision.
8. Production readiness	Assess security, operations, resilience, monitoring, change control, and support.	Signed production-readiness decision; runbooks; rollback; ownership model.

Gate discipline: A failed phase gate is an implementation result. It should generate a remediation plan, revised scope, or stop decision. It should not be converted into a positive performance claim.

5. Governance, roles, and decision rights

OGARD requires technical and operational ownership. The same person may hold several roles in a small pilot, but decision rights should remain explicit. Production implementations should avoid combining responsibilities in a way that lets one person define a rule, approve it, deploy it, and resolve exceptions without review.

Table 3. Core roles and decision rights.

Role	Primary responsibility	Decision rights
Executive or business sponsor	Confirms purpose, priority, resources, risk tolerance, and organizational boundaries.	Approves charter, scope, major risk acceptance, and continuation.
Asset-management owner	Defines asset classes, functional hierarchy, criticality, and domain interpretation.	Approves asset semantics and domain rules.
Maintenance or field subject-matter expert (SME)	Interprets work orders, failure history, maintenance state, and operational context.	Reviews ambiguous maintenance evidence and status rules.
SCADA or historian SME	Interprets tags, telemetry context, tag drift, and source behavior.	Approves tag relationships and telemetry-use constraints.
Data engineering lead	Owns ingestion, conformance, canonical implementation, reconciliation, and technical delivery.	Approves technical design within operator governance.
Data governance lead	Defines ownership, lineage, retention, data quality, metadata, and stewardship.	Approves data contracts and governance records.
Cybersecurity or OT security lead	Assesses access, connectivity, credentials, segmentation, logging, and threat exposure.	Approves security architecture and source access.
Review authority or review group	Accepts, rejects, defers, escalates, or marks contradictions.	Makes authorized mapping and evidence decisions.
Platform or SRE lead	Owns monitoring, deployment, incidents, recovery, and operational support.	Approves operational readiness and runbooks.
Model or AI risk owner	Governs any optional AI task, model, data use, evaluation, and retirement.	Approves or prohibits AI-assisted functions.

5.1. Decision-rights register

Before technical build, create a decision-rights register identifying who may approve source access, accept a mapping, change a rule, change a threshold, resolve a contradiction, approve a derived-status rule, authorize AI use, and approve production use. Each decision should have a named owner and an escalation path.

5.2. Separation of duties

At minimum, production changes to high-impact rules or thresholds should be reviewed by someone other than the original author. Emergency changes should be time-bounded, logged, and retrospectively reviewed. The operator should define which asset classes or outputs require dual approval.

6. Readiness assessment and pilot definition

6.1. Readiness dimensions

Readiness should be assessed at both the source level and the governance level. A technically accessible source may still be unsuitable if ownership, allowed use, time semantics, or security classification are unclear.

Table 4. Source and governance readiness scale.

Score	Source readiness	Governance readiness
0	Unavailable or access prohibited.	No owner, authority, or approved use.
1	Available but unprofiled; identifiers and time fields unknown.	Informal interest only; no approved charter or roles.
2	Profiled; major quality issues identified; extraction not stable.	Owners identified; scope and decision rights incomplete.
3	Usable for conformance with documented limitations.	Pilot charter, source ownership, security review, and review authority approved.
4	Usable for canonical mapping and controlled reconciliation.	Rules, thresholds, data handling, and measurement approach approved.
5	Usable for operator-governed derived status within the approved scope.	Operational ownership, monitoring, change control, and use constraints approved.

A pilot may begin when the governance readiness is at least level 3, one asset class has a named operational owner, and at least two complementary sources are usable for conformance. A stronger pilot normally includes an asset register or functional hierarchy, a maintenance or work-order source, and an operational, fault, inspection, or documentation source.

6.2. Pilot definition

The pilot charter should state the business question in data terms. For example: can the operator reconcile transformer functional locations, installed units, and telemetry references across three named sources while preserving contradictions and review decisions? It should not state an outcome that cannot be attributed to the pilot, such as reducing outages.

Table 5. Pilot go/no-go criteria.

Criterion	Go condition	Stop or remediate condition
Authority	Sponsor, source owners, review authority, and security owner approve the charter.	No named owner or uncertainty about permitted use.
Scope	One asset class, bounded estate, defined time window, named source systems.	Enterprise-wide or multi-domain scope without a controlled learning objective.
Data access	Approved extraction method and credentials; source systems remain authoritative.	Unapproved scraping, uncontrolled credentials, or direct OT writeback.
Evidence diversity	At least two complementary evidence classes and known identity fields.	Only one source or no ability to test cross-system identity.
Review capacity	Authorized SMEs can review a defined volume of ambiguous records.	No review owner or no time allocated for review.
Measurement	Baseline and metric definitions agreed before the pilot run.	Metrics invented after results are known.
Security	Data classification, storage, access, logging, and retention approved.	Protected data would enter an uncontrolled environment.

7. Source discovery and data contracts

7.1. Source-system discovery

The discovery stage should inventory systems that describe, observe, or contextualize the selected asset class. Candidate sources include a computerized maintenance management system (CMMS) or enterprise asset management (EAM) system, an asset register, a geographic information system (GIS) or network model, SCADA or historian references, outage or fault records, inspection and test records, vendor information, technical documents, and planned-outage or resource data. Inclusion should be justified by the pilot question, not by availability alone.

Table 6. Minimum source inventory fields.

Field	Purpose
Source name and type	Identifies the system and evidence class.
Business and technical owner	Establishes authority, stewardship, and escalation.
Approved use	States the pilot purpose and prohibited uses.
Access method	Documents interface, schedule, credentials, network path, and read/write constraints.
Security classification	Defines handling, storage, masking, and retention requirements.
Primary identifiers	Records fields that may represent functional location, equipment, telemetry, or documents.
Time fields and semantics	Distinguishes event time, effective time, extraction time, and ingestion time.
Status fields	Records values, definitions, owners, update behavior, and known limitations.
Refresh and latency	Defines expected frequency, delay, and late-arriving behavior.
Data-quality limitations	Records nulls, duplicates, stale data, drift, undocumented values, and historical gaps.
Target layer	States whether the source contributes raw evidence, conformance, mapping, contradiction, or output context.

7.2. Source data contracts

A source contract should make delivery and interpretation expectations explicit. It is not limited to schema. It should state the meaning of identifiers and time fields, allowed transformations, quality thresholds, failure behavior, security controls, and who is notified when the contract is breached.

Table 7. Minimum source data contract.

Contract area	Minimum content
Delivery	Interface, format, cadence, expected volume, backfill method, and service window.
Schema	Fields, types, allowed values, nullable fields, version, and change-notification process.
Identity	Identifier meaning, uniqueness scope, aliases, replacement behavior, and known reuse.
Time	Event, effective, extraction, ingestion, and update timestamps with time zone and precision.
Quality	Required fields, duplicate policy, stale-data threshold, referential checks, and quarantine behavior.
Security	Classification, encryption, access, credential handling, logging, retention, and deletion.
Ownership	Business owner, technical owner, steward, support contact, and escalation path.
Change management	Notice period, compatibility expectation, test data, and rollback or replay procedure.

Interpretation rule: A source status field is evidence from that source. It is not automatically the canonical or operator-approved status.

8. Identifier-regime mapping

The identifier-regime map is the core discovery artifact. It classifies each identifier by what it represents, its scope, its stability, and its behavior over time. The three primary regimes are functional location, physical equipment identity, and telemetry or instrumentation identity. Additional identifiers, such as work-order numbers, document references, GIS features, or outage-event identifiers, are evidence links rather than substitutes for these regimes.

Table 8. Identifier-regime mapping worksheet.

Field	Example question
Source and field	Which source and column or attribute contains the identifier?
Regime	Does it represent a functional role, physical item, telemetry channel, or evidence record?
Uniqueness scope	Is it unique globally, by source, by site, by asset class, or only within a time period?
Time behavior	Does it persist through equipment replacement, relocation, retagging, or network reconfiguration?
Alias patterns	Are there approved abbreviations, prefixes, suffixes, or legacy values?
Reuse risk	Can the identifier be reused for a new item or repointed to a different location?
Authority	Who can confirm or correct the relationship?
Mapping evidence	Which records, dates, documents, or rules can support the relationship?
Review condition	When must the relationship be routed to review?

8.1. Time-valid relationships

Relationships should carry valid-from and valid-to values where changes are possible. A functional location can remain stable while equipment is replaced. A physical unit can move. A telemetry tag can be renamed, repointed, or left stale. The implementation should avoid a permanent one-to-one relationship unless the operator confirms that the source semantics justify it.

8.2. Unknown and unresolved identity

An unresolved identity is a legitimate output. The implementation should use an explicit unresolved or unknown state rather than forcing a candidate mapping. The pilot report should measure unresolved volume and explain the evidence gaps that produced it.

9. Canonical asset-and-evidence implementation

The canonical model should represent the governed asset context and preserve the evidence that supports, challenges, or contextualizes it. Selected IEC CIM concepts can inform the model, but the implementation should describe this as mapping guidance unless formal profile conformance is separately defined and tested [2], [3].

Versioning requirement. Every persisted canonical, evidence, mapping, review, contradiction, derived-status, and audit record must carry, or be traceable to, both `model_profile_version` and `schema_version`. Historical outputs must remain traceable to the model and schema in force when the decision was produced; later model changes must not silently reinterpret prior records.

Table 9. Minimum canonical and evidence entities.

Entity	Minimum role
CanonicalAsset	Stable governed reference context for an asset within the selected implementation scope.
FunctionalLocation	Role or position in the network or operating hierarchy, with time validity where applicable.
EquipmentUnit	Physical installed item with serial or alternative identity and lifecycle history where available.
AssetInstallation	Time-valid relationship between a functional location and an equipment unit.
TelemetryChannel	Observation channel, source historian, tag, measurement type and unit, and time-valid relationship to a location, item, or aggregate.
EvidenceRecord	Preserved source record with source, identifier, timestamps, batch or stream identifier, security classification, schema version, quality state, and original values or payload reference.
EvidenceRecord subtype examples	Typed evidence for maintenance or work orders, faults or outages, inspections or condition records, and technical documents.
MappingDecision	Candidate context, method tier, rule or model version, score or confidence, threshold, outcome, actor, reason code, time, effective validity period, and evidence references.
ReviewAction	Authorized acceptance, rejection, deferral, escalation, or contradiction decision with reason and evidence.
Contradiction	Typed disagreement, severity, implicated records, owner, state, and resolution.
DerivedAssetStatus	Operator-governed status, evidence set, effective interval, rule version, review state, expiry, and contradiction references.

9.1. Source evidence is immutable in meaning

The implementation may store technical copies and may correct ingestion defects, but it should not rewrite the meaning of the original record. Normalized values should remain linked to the original values and

transformation. Corrections to source data should occur through the source owner or a documented override process, not by silently editing evidence in the canonical layer.

9.2. Canonical values and survivorship

Where an attribute requires a canonical value, survivorship should be defined per attribute and source class. The selected value, losing values, rule, effective date, and contradiction state should be retained. A golden record without the disagreement history is insufficient for consequential asset evidence.

10. Layered data architecture and ingestion controls

The reference architecture uses four logical processing layers: raw evidence, conformance, canonical asset-and-evidence, and consumption. The source interface is an approved access boundary rather than a data layer. Review, contradiction, and audit are cross-cutting control functions that operate across the canonical and consumption stages.

Table 10. Architecture layers and implementation guarantees.

Architecture element	Primary work	Required guarantee
Source interface boundary	Approved read, extract, file, API, event, or controlled replication method.	Access and delivery are authorized, logged, and bounded.
Raw evidence layer	Preserve source payload or reference, source identifier, extraction and ingestion time, batch or stream identifier, and schema version.	The original evidence used by the implementation can be reconstructed.
Conformance layer	Validate schema, normalize controlled fields, standardize time, handle technical duplicates, and quarantine invalid records.	Records are comparable and their quality state is explicit without erasing source evidence.
Canonical asset-and-evidence layer	Create time-valid asset relationships, mappings, review outcomes, evidence sets, contradictions, and status provenance.	Every accepted relationship and derived output has traceable evidence and decision history.
Consumption layer	Expose evidence views, review queues, contradictions, operator-approved status, and measurement tables.	Consumers receive only governed outputs allowed for their role and use case, with lineage and review state.

10.1. Ingestion requirements

- Loads should be idempotent or otherwise protected against unintended duplication.
- Failed records should be retained, quarantined, or rejected under documented severity rules.
- Late-arriving and out-of-order records should preserve event and effective time rather than being interpreted only by arrival time.
- Schema drift should trigger controlled handling, alerting, and contract review.
- Replay and backfill procedures should avoid changing historical decisions without a versioned recomputation record.
- Source deletion or correction events should be represented according to source semantics and retention policy.

10.2. Consumption restrictions

A consumption view should state its intended purpose, freshness, coverage, quality limitations, review state, and prohibited uses. Pilot outputs should be labeled as pilot outputs. Unreviewed candidates should not appear indistinguishably from accepted mappings. Unknown or stale status should remain visible.

11. Reconciliation rules, scoring, and calibration

11.1. Candidate generation and blocking

Candidate generation should restrict comparison to plausible relationships. Blocking may use region, station, voltage class, equipment class, identifier fragments, functional hierarchy, installation windows, document references, or approved source relationships. Blocking rules should be versioned and evaluated for the risk of excluding true candidates.

11.2. Deterministic rules

Deterministic rules should be used where the relationship is sufficiently defined and testable. Examples include exact serial matches within an approved scope, validated alias tables, installation records, approved telemetry naming conventions, or direct source relationships. Exact text equality is not automatically authoritative if the source permits reuse or drift.

Table 11. Minimum reconciliation rule record.

Field	Minimum content
Rule identifier and version	Stable identifier, semantic version, effective date, and release reference.
Purpose and owner	What the rule resolves, asset or source scope, and accountable owner.
Inputs	Source systems, fields, required values, time fields, and quality prerequisites.
Condition	Explicit match logic, exclusions, and temporal constraints.
Outcome	Auto-accept, candidate for review, reject, defer or unresolved, or contradiction (Table 12).
Risk class	Operational consequence and approval level for a false positive or false negative.
Tests	Positive, negative, boundary, temporal, and regression cases.
Monitoring	Volume, acceptance rate, error signal, drift signal, and review feedback.
Change control	Approver, reason, prior version, deployment record, and rollback.

11.3. Similarity-based scoring

Similarity scoring should be reproducible, feature-limited, and calibrated on operator-labeled examples. Candidate features may include token similarity, Jaro–Winkler similarity, character n-gram similarity, approved domain tokens, time windows, and constrained semantic re-ranking. The score should not be treated as a probability unless it has been calibrated and validated as such.

11.4. Threshold calibration

Thresholds should be set using a labeled calibration sample that is separate from any evaluation sample used to report performance. Thresholds may vary by asset class, source quality, and consequence. A high-impact mapping should normally require stronger evidence or review than a low-consequence documentation link.

11.5. Cold-start calibration

A first pilot may have no reliable labeled calibration sample. In that situation, similarity-based candidates should not be automatically accepted. The implementation should begin in review-first mode: deterministic rules may operate only where their authority is already documented, while similarity candidates are routed to authorized review and their outcomes are recorded as the initial labeled set.

After a sufficient and representative set of review decisions exists, the team should separate calibration and evaluation samples, document class balance and exclusions, set conservative thresholds, and re-baseline review volume before enabling any limited automatic acceptance. Historical review outcomes should not be assumed correct without quality checks, and the initial threshold profile should remain versioned and reversible.

Table 12. Reference routing outcomes.

Outcome	When appropriate	Required record
Auto-accept	A versioned deterministic rule or approved calibrated threshold satisfies the evidence and risk requirements.	Rule, inputs, version, score if used, evidence references, and time.
Candidate for review	Evidence is plausible but incomplete, ambiguous, or consequential.	Candidate set, scores, rationale, contradictions, and review owner.
Reject	Evidence is incompatible, below threshold, outside time validity, or prohibited by a rule.	Reason code, rejected candidate, rule or score, and evidence.
Defer or unresolved	Available evidence cannot support acceptance or rejection.	Evidence gap, owner, next action, and review date.
Contradiction	Sources or relationships disagree materially.	Contradiction type, severity, implicated records, and routing.

Scoring scope: A routing rule considers the score alongside source records, time context, and operator authority.

12. Human review and audit workflow

Human review is a core workflow. It should present enough evidence for an authorized reviewer to make a bounded decision without searching across every source manually. Review volume should be designed and measured as part of the pilot.

Table 13. Minimum human review record.

Field	Minimum content
Review item	Unique identifier, creation time, priority, age, and owning queue.
Source evidence	Records, original identifiers, relevant values, timestamps, and quality state.
Candidate context	Candidate asset, location, equipment, telemetry, or document relationships.
Method context	Rules applied, similarity features, scores, thresholds, and model task if any.
Contradictions	Known disagreements, stale evidence, missing evidence, and time ambiguity.
Decision options	Accept, reject, defer, escalate, mark contradiction, or create rule candidate.
Reviewer	Identity, role, authority, decision, reason code, time, and comments.
Follow-up	Rule change, source correction, additional evidence, owner, and due date.

12.1. Queue design

Queues may be separated by asset class, source type, consequence, contradiction severity, or reviewer expertise. The implementation should monitor queue volume, age, completion time, decision distribution, and repeated patterns. A queue that grows without ownership is a production defect.

12.2. Review feedback

Repeated approved decisions may justify a new deterministic rule or threshold change. Promotion should require documented evidence, tests, approval, and versioning. A reviewer decision should not automatically train or alter a model or rule library without governance.

12.3. Review-capacity planning

Review ownership should be converted into a capacity estimate before the pilot begins. A simple planning calculation is: $\text{expected review hours per period} = \text{reconciliation-eligible records per period} \times \text{expected review-routing share} \times \text{average review minutes per item} \div 60$. The estimate should be calculated for the primary threshold profile and at least one more conservative profile.

The pilot plan should record expected record volume, estimated routing share, reviewer availability, target queue age, escalation path, and stop or scope-reduction criteria. Actual routing volume, completion time, unresolved volume, and repeat-review patterns should be reported during the pilot. A threshold profile that produces a queue the authorized team cannot clear is not operationally feasible even when its mapping metrics appear strong.

13. Contradiction handling and derived status

13.1. Contradictions

Contradictions are preserved because disagreement can be operationally meaningful. A register may show active while telemetry is stale. A serial may appear at two locations. A work order may refer to equipment that was replaced before the event. An outage source may show offline while another source remains unchanged. The implementation should type, prioritize, route, and resolve contradictions rather than silently selecting a winner.

Table 14. Minimum contradiction record.

Field	Minimum content
Contradiction identifier	Stable identifier and creation time.
Type and severity	Operator-approved taxonomy, consequence, and priority.
Asset context	Functional location, equipment, telemetry, or evidence context affected.
Implicated records	Source identifiers, timestamps, values, and quality state.
Detection method	Rule, comparison, threshold, review action, or manual report.
Owner and route	Assigned team, queue, escalation, and target response.
Resolution	Accepted evidence, source correction, rule change, waiver, or unresolved outcome.
Audit	Resolver, authority, time, rationale, and linked change records.

13.2. Operator-governed derived status

Derived status is an information product created under operator-approved rules from a defined evidence set. It records a rule-based interpretation of the supporting evidence. Each rule should define required evidence classes, freshness, exclusions, contradiction behavior, review requirement, effective time, expiry, and owner.

Table 15. Reference derived-status taxonomy.

Status	Reference meaning	Minimum interpretation safeguard
Available	Approved evidence supports expected availability within the defined scope, and no unresolved material contradiction blocks the classification.	Availability is assessed within the defined evidence scope. The operator defines coverage, freshness, and disqualifying contradictions; latent defects require separate assessment.
Degraded	Approved evidence supports impaired condition or reduced capability under an operator-defined rule.	Requires asset-class-specific criteria, time validity, and authorized engineering interpretation.
Offline	Approved evidence supports that the asset is unavailable, disconnected, or out of service.	Must distinguish planned, forced, temporary, and data-quality conditions using time-valid evidence.
At risk	Approved evidence indicates a condition requiring attention but does not support Degraded or Offline.	Must not be represented as a failure prediction; indicators, expiry, and review requirements must be defined.
Unknown	Evidence is insufficient, stale, ambiguous, outside scope, or materially contradictory.	Unknown is the safe default whenever another status cannot be justified.

13.3. Diagnostic-support information

The implementation may assemble maintenance history, telemetry context, fault records, documents, inspections, and prior review decisions to support operator-led investigation. It should not label a proposed cause as a diagnosis unless the operator has an approved process and an authorized person has made that determination. Optional summaries should cite the evidence used.

14. Optional AI-assisted functions

Artificial intelligence is optional. An implementation that does not use AI can still satisfy OGARD. Where AI is used, it should be governed as an advisory component under the principle that it may recommend,

summarize, or prioritize but may not become the authority. NIST AI RMF 1.0 organizes AI risk-management activity around govern, map, measure, and manage functions [9]. NIST AI 600-1 is a companion profile of the AI RMF for generative AI [10].

Table 16. AI-assisted function control matrix.

Function	Permitted role	Prohibited role	Minimum control
Candidate mapping	Propose or re-rank bounded candidates with rationale and evidence links.	Create a final mapping without an approved rule or review.	Task version, model version, candidate bounds, confidence, audit, and review.
Evidence summarization	Summarize selected records for a reviewer.	Invent facts, omit material contradictory evidence, or replace source review.	Source citations, prompt or task record, output retention, and reviewer confirmation.
Document classification	Suggest type, applicability, or routing.	Change canonical attributes or status automatically.	Approved label set, evaluation sample, confidence, and exception path.
Contradiction triage	Suggest category or priority for a queue.	Suppress or resolve a contradiction autonomously.	Rule boundary, evidence links, reviewer override, and monitoring.
Explanation drafting	Draft a human-readable explanation of a rule or evidence set.	Create unsupported causal conclusions.	Traceability, prohibited-claim tests, and review.

14.1. AI use-case approval record

Before enabling an AI task, document the purpose, users, data classes, model and version, deployment boundary, inputs, outputs, evaluation method, risk owner, prohibited uses, human review, retention, monitoring, and retirement conditions. Sensitive operator data should not be sent to an external service unless the operator has explicitly approved the service, contract, architecture, and data handling.

14.2. Evaluation

Evaluate the AI task on a representative, approved sample that includes ambiguous and contradictory cases. Report failure types, not only aggregate accuracy. A model that produces fluent but unsupported explanations is unsuitable even when classification metrics appear acceptable. Any model or prompt change should trigger versioned re-evaluation.

15. Security, confidentiality, and OT boundary

Security controls should be tailored to the operator environment and risk. NIST CSF 2.0 emphasizes governance alongside identify, protect, detect, respond, and recover functions [5]. NIST SP 800-53 provides a customizable security and privacy control catalog [7], and CISA publishes prioritized performance goals for critical infrastructure [8]. OGARD does not replace those frameworks or operator requirements.

Table 17. Minimum security and confidentiality controls.

Control area	Minimum expectation
Architecture boundary	Document IT, OT, vendor, cloud, and public-release trust boundaries. Default to no direct operational control or writeback.
Access control	Least privilege, role-based access, named accounts, periodic review, and separation of privileged duties.
Credentials and secrets	Approved secret storage, rotation, no hard-coded secrets, and access logging.
Encryption	Approved protection in transit and at rest according to classification and operator policy.
Network and source access	Operator-approved interfaces, segmentation, monitoring, and constraints on OT connectivity.
Audit and monitoring	Log source access, administrative actions, deployments, rule changes, review decisions, and data exports.
Data minimization	Ingest only fields and history needed for the approved purpose. Mask or exclude unnecessary protected data.
Retention and deletion	Document retention, legal hold, archive, disposal, and deletion verification.
Environment separation	Separate development, test, production, and public reference materials. No production data in public artifacts.
Incident response	Define notification, containment, evidence preservation, recovery, and post-incident review.
Third-party and supply chain	Assess dependencies, libraries, services, licenses, vulnerabilities, and provider data use.

15.1. Public and operator artifact separation

Public documentation, synthetic data, example configurations, and reference code should be maintained separately from operator repositories and environments. The public project should not contain live asset records, screenshots of protected systems, employer code, client documents, credentials, network details, or non-public operating information.

15.2. OT interaction

A pilot should normally consume approved copies or read-only interfaces rather than creating direct control paths. Any connection to OT should be assessed and approved by the operator security and operations functions. Data freshness requirements should not be used to justify bypassing segmentation, change control, or safety practices.

16. Testing, quality assurance, and observability

Testing should cover data, rules, time behavior, review workflows, security, failure handling, and operational support. A successful pipeline run does not demonstrate that mappings or status semantics are correct.

Table 18. Test categories and evidence.

Category	Examples	Evidence retained
Unit and schema	Parsing, required fields, types, identifier normalization, and rule functions.	Automated test results and version.
Data quality	Nulls, duplicates, stale data, range, format, referential integrity, and quarantine.	Quality report and exception samples.
Temporal	Replacement windows, late arrival, out-of-order events, effective time, and expiry.	Time-based test cases and expected outcomes.
Reconciliation	Positive, negative, ambiguous, cross-regime, and blocked-candidate cases.	Labeled cases, results, errors, and review routes.
Contradiction	Status disagreement, duplicate location, stale tag, and time conflict.	Detected and missed cases with reason.
Derived status	Required evidence, stale evidence, disqualifying contradiction, Unknown fallback, and expiry.	Rule test suite and operator approval.
Review workflow	Assignment, authorization, escalation, audit, queue aging, and rule promotion.	Workflow logs and reviewer acceptance.
Security	Access, secrets, logging, data export, vulnerability, dependency, and failure cases.	Security test or assessment record.
Resilience	Retry, replay, source outage, schema drift, rollback, recovery, and dependency failure.	Runbook exercise and recovery evidence.
Performance	Volume, latency, queue growth, storage, and peak load within the pilot scope.	Benchmark with environment and limitations.

16.1. Observability

Monitor pipeline health, source freshness, schema changes, conformance failures, quarantine volume, candidate volume, acceptance and rejection distribution, queue age, unresolved contradictions, rule errors, status expiry, audit completeness, and downstream availability. Alerts should identify an owner and an expected response.

16.2. Data and rule drift

Drift may arise from identifier changes, new asset classes, source upgrades, status-value changes, retagging, replacement behavior, or model changes. The implementation should detect and review drift before it silently changes mapping or status outcomes.

17. Measurement and pilot reporting

17.1. Separate evidence classes

Table 19. Measurement classes and claim boundaries.

Class	Appropriate measures	Claims not supported
Public release	Document completeness, artifact inventory, reproducibility instructions, issue tracking, reviewer comments, and version history.	Operator performance, adoption, or live reliability benefit.
Synthetic benchmark	Estate size, scenario coverage, per-tier routing, precision and recall against seeded truth, unresolved review, contradiction detection, and reproducibility.	Live utility performance, outage reduction, or expected operator results.
Operator pilot	Source coverage, asset coverage, mapping quality on approved samples, review burden, contradiction handling, evidence assembly time, audit completeness, and user feedback.	Production-scale or causal reliability impact unless separately designed and validated.
Production operation	Service reliability, sustained data quality, review performance, incident history, rule drift, and operator-defined outcome measures.	Causal outage or restoration impact without an appropriate study design and baseline.

17.2. Baseline and measurement plan

Metrics should be defined before results are known. The plan should state the population, sampling method, time period, source, numerator and denominator, exclusions, reviewer process, uncertainty, and claim boundary. Where no reliable baseline exists, the pilot should report activity and quality measures rather than inventing an improvement percentage.

Table 20. Pilot metric dictionary.

Metric	Reference definition	Minimum reporting context
Source coverage	Included sources divided by the sources defined as necessary for the pilot scope.	Scope, excluded sources, and rationale.
Asset coverage	Assets with sufficient evidence for the stated output divided by in-scope assets.	Asset class, evidence requirement, and unknown volume.
Accepted mapping precision	Correct accepted mappings divided by reviewed accepted mappings in an approved sample.	Sampling method, reviewer authority, confidence interval where appropriate.
Review volume	Records or relationships routed to review.	Total population, routing rules, and repeat items.
Unresolved volume	Review items remaining unresolved at the reporting cutoff.	Age, reason codes, and owner.
Contradiction detection	Known or reviewed contradictions surfaced by the implementation.	Ground truth or review method and scenario scope.
Audit completeness	Material decisions containing all required audit fields.	Required field definition and sampled population.

Metric	Reference definition	Minimum reporting context
Evidence assembly time	Elapsed time to produce the defined evidence package under the pilot workflow.	Start and stop events, comparison method, and user role.
Reviewer throughput	Completed authorized review actions per defined period.	Queue type, staffing, complexity, and quality check.

17.3. Pilot report

The pilot report should state the problem, scope, environment, data type, sources, asset class, owners, methods, rule versions, thresholds, review process, measurements, errors, unresolved issues, security limitations, user feedback, and recommended next actions. It should identify which findings are technical, operational, or organizational. It should state clearly what was not measured.

18. Production readiness and operations

A pilot should not move into production because it demonstrated a technically interesting result. Production readiness requires durable ownership, approved use, security, monitoring, incident response, change control, review capacity, support, and a safe failure mode.

Table 21. Production-readiness gate.

Area	Required evidence before approval
Purpose and scope	Approved use cases, users, prohibited uses, asset classes, source systems, and output definitions.
Ownership	Named business, data, security, review, platform, and support owners.
Data governance	Approved contracts, lineage, retention, quality controls, source correction, and stewardship.
Security	Approved architecture, access, secrets, network, monitoring, dependencies, and incident response.
Reconciliation	Versioned rules, calibrated thresholds, testing, error analysis, and controlled change process.
Review operation	Authorized reviewers, queues, service levels, escalation, quality assurance, and capacity.
Derived status	Operator-approved definitions, evidence, contradiction behavior, expiry, and fallback to Unknown.
AI functions	Approved use-case records, evaluation, human review, monitoring, and ability to disable.
Reliability and recovery	Runbooks, rollback, replay, backup, recovery objectives, source-outage behavior, and exercises.
Monitoring	Dashboards, alerts, owners, thresholds, audit completeness, and drift detection.
User readiness	Training, documented limitations, feedback route, and acceptance.
Legal and regulatory	Operator review of applicable contractual, records, privacy, safety, and regulatory obligations.

18.1. Safe failure behavior

When evidence is stale, access fails, rules cannot execute, audit records are incomplete, or a material contradiction is unresolved, the system should fail safely. Depending on the use case, this may mean marking

the output Unknown, withholding it, routing it to review, or reverting to the operator-approved prior process. It should not silently preserve a stale status as current.

18.2. Change management

Schemas, source mappings, rules, thresholds, derived-status definitions, AI tasks, dependencies, and measurement definitions should be versioned. Changes should record reason, owner, approval, tests, effective date, deployment, monitoring period, and rollback. Material changes may require recalibration or renewed operator acceptance.

18.3. Operations and periodic review

The operator should schedule periodic review of source ownership, access, rules, thresholds, review decisions, contradictions, security posture, AI use, retention, performance, and the continued need for each output. Retirement should be planned for sources, rules, models, and the implementation itself.

19. Scope and evaluation limits

This guide defines an implementation method for operator-led pilots. Live operator validation is a future development stage. Each deployment requires evaluation for its asset class, source systems, architecture, security environment, and intended use.

- Operator adoption and production deployment would be documented separately for each implementation.
- Outage reduction, restoration time, maintenance optimization, and reliability outcomes require operator records and a defined evaluation design.
- Asset condition, fault diagnosis, and operational actions remain operator-governed decisions.
- Formal CIM conformance, NERC CIP compliance, regulatory approval, and cybersecurity certification follow their respective assessment processes.
- Synthetic or pilot results should not be generalized beyond their documented scope.
- The quality of outputs depends on source quality, identity history, domain expertise, governance, review capacity, and implementation discipline.
- AI-assisted functions are optional and may be prohibited by an operator.
- The guide does not replace engineering, safety, cybersecurity, legal, privacy, records, or operational procedures.

The intended contribution is a transparent implementation method that makes evidence, assumptions, uncertainty, decisions, and limitations inspectable. A responsible implementation should preserve the ability to say unknown and to stop when the available evidence or governance does not justify a stronger output.

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Appendix A. Minimum implementation records

Table 22. Minimum implementation records.

Record	Minimum fields
Pilot charter	Problem, purpose, asset class, sources, estate, time window, users, owners, decision rights, security, success criteria, exclusions, and evaluation boundaries.
Readiness assessment	Source and governance scores, evidence, issues, remediation, owner, and go/no-go decision.
Source inventory	Fields listed in Table 6, source approval, and version.
Source data contract	Fields listed in Table 7, approvers, effective date, and change history.
Identifier-regime map	Fields listed in Table 8.
Canonical schema record	Entity, attribute, type, definition, source, time behavior, sensitivity, owner, and mapping guidance.
Reconciliation rule	Fields listed in Table 11, tests, approval, deployment, monitoring, and retirement.
Threshold calibration	Labeled sample, split method, features, thresholds, errors, risk class, approval, and limitations.
Human review record	Fields listed in Table 13 and linked evidence.
Contradiction record	Fields listed in Table 14 and resolution history.
Derived-status rule	Status, scope, required evidence, freshness, exclusions, contradiction behavior, review, expiry, owner, and tests.
AI use-case record	Purpose, model, version, data, task, evaluation, risks, human control, monitoring, and retirement.
Audit event	Event, actor, action, object, source references, rule or model, version, time, reason, and outcome.
Pilot measurement plan	Metric, definition, baseline, population, sample, period, source, exclusions, uncertainty, and claim boundary.
Pilot report	Scope, environment, sources, method, results, errors, unresolved issues, security, limitations, decisions, and next actions.
Production-readiness record	Table 21 evidence, approvers, conditions, residual risks, and decision.

Every persisted canonical, evidence, mapping, review, contradiction, derived-status, and audit record carries, or is traceable to, `model_profile_version` and `schema_version` (Section 9).

Appendix B. Phase-gate checklist

B.1. Before data access

- ☐ Pilot purpose and direct outputs are defined in data terms.
- ☐ Asset class, source systems, estate boundary, and time window are documented.
- ☐ Business, source, security, data, and review owners are named.
- ☐ Approved use, prohibited use, classification, retention, and public-artifact boundary are documented.
- ☐ No uncontrolled OT writeback or unapproved external service is included.
- ☐ Measurement definitions and evaluation boundaries are agreed.

B.2. Before reconciliation

- ☐ Raw source evidence is preserved with source and time metadata.
- ☐ Conformance transformations and quality state are traceable.
- ☐ Identifier regimes and time-valid relationships are documented.
- ☐ Candidate generation and deterministic rules are versioned and tested.
- ☐ Similarity features are reproducible; thresholds are calibrated on a labeled sample or, in a cold start (Section 11.5), similarity candidates are routed to authorized review.
- ☐ Review authority, queue, decision options, and audit fields are functioning.

B.3. Before derived status

- ☐ Status terms are approved by the operator for the selected asset class and use case.
- ☐ Required evidence, freshness, exclusions, contradiction behavior, review, and expiry are documented.
- ☐ Unknown is the safe outcome when another status cannot be justified.
- ☐ Rules have positive, negative, stale, contradictory, and boundary tests.
- ☐ Consumers understand that the output is an operator-governed information product.

B.4. Before production consideration

- ☐ Pilot report and error analysis are complete.
- ☐ Security, access, monitoring, incident response, rollback, replay, and recovery are approved.
- ☐ Review capacity and service levels are sustainable.
- ☐ Change control covers sources, schemas, rules, thresholds, status definitions, and AI tasks.
- ☐ Training, support, documentation, periodic review, and retirement are assigned.
- ☐ Residual risks and production conditions are accepted by authorized operator owners.

Appendix C. Example bounded pilot

This illustrative synthetic example shows the scope and records a bounded pilot would use.

Table 23. Example pilot charter summary.

Field	Synthetic example
Question	Can transformer functional locations, installed equipment units, and telemetry references be reconciled across three source classes while preserving contradictions and review decisions?
Asset class	Primary distribution transformers in one synthetic region.
Sources	Synthetic asset register, CMMS or work-order records, and telemetry-reference history, with a small inspection-evidence set.
Time window	Two years including several synthetic equipment replacements and retagging events.
Direct outputs	Identity relationships, unresolved candidates, contradiction records, audit records, and an evidence view.
Not in scope	Live SCADA connection, outage prediction, autonomous diagnosis, operational writeback, and reliability-outcome claims.
Review roles	Synthetic asset-management, maintenance, and telemetry reviewers.
Measures	Coverage, reviewed precision, unresolved volume, contradiction handling, audit completeness, and reviewer workload.
Exit decision	Proceed, remediate and repeat, narrow scope, or stop.

Appendix D. Glossary

Asset context. The governed representation assembled for an asset situation, distinct from the source evidence that describes it.

Audit record. A record that identifies an event, actor, action, evidence, rule or model, version, time, reason, and outcome.

Blocking. Restriction of candidate generation to plausible comparison sets before scoring.

Canonical model. A governed representation used to relate asset context and evidence across sources.

Contradiction. A typed disagreement between evidence classes, identifiers, statuses, or time contexts.

Derived status. An operator-governed information product produced under a documented rule from a defined evidence set.

Evidence record. A preserved source record, event, observation, document reference, or transaction.

Functional location. The role or position occupied in the network or operating hierarchy.

Method tier. The reconciliation method that produced a route: Tier 1, deterministic rules; Tier 2, reproducible similarity scoring; Tier 3, optional AI-assisted recommendation (advisory only). Authorized human review decides unresolved or consequential cases.

Physical equipment identity. The installed item, commonly associated with serial, manufacturer, nameplate, and installation history.

Review authority. A person or group authorized to accept, reject, defer, escalate, or resolve a material implementation decision.

Telemetry identity. The observation channel or tag through which operating data is recorded.

Unknown. The safe output when available evidence does not justify another status or relationship.